SE423 Software Project Management

Artifact 2: Methodology Analysis, Risks, and Decision

**Scenario 11: National Employment and Skill Matching Platform**

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**1. Agile 1**

1.2 Match with Team/Organizational Structure & Stakeholders

**2. Agile 2**

2.1 Strengths & Weakness

2.2 Match with Team/Organizational Structure & Stakeholders

**3. Plan-Driven 1**

3.1 Strengths & Weakness

3.2 Match with Team/Organizational Structure & Stakeholders

**4. Plan-Driven 2**

4.1 Strengths & Weakness

*Strengths*

1. Proper Requirements and Documentation: Plan-driven development process stresses on the importance of requirement definition and documentation at the beginning of the software development process. It will be easy to understand how the functions of Employer Portal are expected to work, including the features such as employer's profile, job postings, applications, filtering of candidates, and conducting interviews.
2. Predictable Scheduling and Budgeting: The tasks planned in advance make it possible to assess the necessary resources, budget, and schedule. As a result, the development process will fit within project constraints.
3. Defined Responsibilities: Team members' responsibilities can be properly defined

during the process of requirement definition, design, implementation, and testing.

4. Quality Assurance: It is through the review process that it will be possible to ensure that all the necessary requirements have been met. In particular, it concerns authentication, application management, and security of candidate information.

*Weaknesses*

1. Inflexibility: Any changes in requirements after the planning stage are likely to be hard and expensive. It is possible that employers will need more screening requirements and changes in the hiring process at the development stage.
2. Lack of Early Feedback from Stakeholders: The stakeholders may only be able to interact with the application in the developed state and may not provide feedback until then. This makes it likely that some issues related to usability will be revealed later on.
3. More Documentation Work: Doing more work to write and maintain documentation adds up to extra work, which could prove to be difficult for a small development team.
4. Potential for Redesigning/Re-development: In case the requirements are not correct, the team will have to redesign or re-develop the already developed components of the system.

4.2 Match with Team/Organizational Structure & Stakeholders

The plan-driven approach helps in organizing the development process but it is not very well-suited for Fursah due to the organizational structure of Fursah and dynamic needs of its stakeholders.

Team Structure: For its first deliverable, Fursah decided to organize a creative team with an emphasis on experiments, diverse skills and flexibility. The plan-driven approach will make it difficult for the company to do this because requirements and processes of

development will be defined beforehand. Even though responsibilities will make employees more accountable, lack of flexibility will make it difficult for Fursah to experiment, especially when developing the smart matching engine through Natural Language Processing (NLP).

Organizational Structure: Organizational structure of Fursah is matrix, which means functional units with project-oriented responsibilities. A plan-driven approach can be helpful in creating schedules, documents and processes within departments of Fursah. But matrix organizational structure has been chosen to respond quickly to the dynamic changes in the job market. Thus, the change control process will make the decision making process very difficult for the organization.

Stakeholders: There are many stakeholders in Fursah who include employees, employers, governments, and education/training institutions. The plan-driven methodology will ensure that the requirements of all these stakeholders are collected and analyzed before implementation of development. This will be helpful for the government integration process, data security, and regulation needs. However, there can be changes in requirements of employers, employees, and market forces during the whole project. Lack of involvement of stakeholders in the development process might lead to implementation of features that no longer satisfy their needs.

Overall Suitability: There are several strengths of the plan-driven methodology such as documentation, predictability and control. However, there are some weaknesses of the methodology for our case that include inability to provide flexibility needed for our team that requires creativity and matrix organizational structure.

## 5. Selected Process Methodology

### 5.1 Discussion On Selected

5.2 Risk Register

SE423 Artifact 2 (Risk Register)